Dongkuan Zhang

Nationality: Chinese

Date of birth: 02.05.1995
Tel: +86 15510689505
Email: dkz@mail.dlut.edu.cn
Location: Dalian, Liaoning, China

EDUCATION

2023.09 – Present PhD Candidate, Environmental Science and Engineering, Dalian University of Technology
2020.09 – 2023.07 Master, Environmental Science and Engineering, College of Life and Environmental Sciences, Minzu University of China
2013.09 – 2017.07 Bachelor, Petroleum Engineering, College of Petroleum and Natural Gas Engineering, Liaoning Shihua University

ACADEMIC ACHIEVEMENTS

Publications:

1. **Dongkuan Zhang**, Tanzila Anjum, Zhiqiang Chu, Jeffrey S. Cross, Guozhao Ji. Simulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integration. Renewable and Sustainable Energy Reviews, 2025, 221: 115895. https://doi.org/10.1016/j.rser.2025.115895
2. **Dongkuan Zhang**, Tong Ren, Yudong Liu, Minglei Li, Junchao Yang, Jiacheng Dai, Jing Li, Jeffrey S. Cross, Guozhao Ji. Clogging mechanism and early-warning criterion for shear-thinning distillation residues in twin-fluid spray nozzles based on gas-liquid multiphase flow. Chemical Engineering Research and Design, 2026, 232: 619–626. https://doi.org/10.1016/j.cherd.2026.07.045
3. **Dongkuan Zhang**, Zhiqiang Chu, Hong Tian, Jeffrey S. Cross, Guozhao Ji. Input convex neural network modeling for rheological performance evaluation of non-Newtonian fluid residues in chemical reactors. Journal of Xi ’ an Jiaotong University, 2025, 59(12): 80-91. https://zkxb.xjtu.edu.cn/zh/article/doi/10.7652/xjtuxb202512007
4. **Dongkuan Zhang**, Meilin Liu, Jianxin Xia. Experimental study on sediment disturbance during collection of deep-sea polymetallic nodules. Mining and Metallurgical Engineering, 2023, 43(3): 20-23. https://doi.org/10.3969/j.issn.0253-6099.2023.03.005
5. **Dongkuan Zhang**, Lei Guan, Huade Cao, Jianxin Xia. Matching of flow field parameters of deep-sea polymetallic nodules fluidized collector. Journal of Ocean Technology, 2022, 41(4): 53-60. http://qikan.cqvip.com/Qikan/Article/Detail?id=7107826763
6. Junchao Yang, **Dongkuan Zhang**, Li Liu, Yuanbo Xie, Yingnan Du, Aimin Li, Changlei Qin, Guozhao Ji. Eulerian-Lagrangian model for simulation of flow behavior and heat transfer of lab scale dual circulating fluidized beds. Applied Thermal Engineering, 2025, 279: 127937. https://doi.org/10.1016/j.applthermaleng.2025.127937
7. Tanzila Anjum, **Dongkuan Zhang**, Gianni Olguin, Bradley Ladewig, David K. Wang, Xiaozhen Zhang, Asim Laeeq Khan, Guozhao Ji; Quantifying the Influence of Supercritical Drying on the Pore Size Distribution of Cobalt-Doped Silica Membrane. *Ind. Eng. Chem. Res.* 1 July 2026; 65 (25): 13372–13381. https://doi.org/10.1021/acs.iecr.6c01111

8. Jiacheng Dai, Yingnan Du, Yuanbo Xie, **Dongkuan Zhang**, Li Liu, Yang Gui, Guozhao Ji. The Eulerian–Lagrangian Model for Simulating the Moisture Content Effect on the Characteristics of MSW Combustion in a 50 T/D Grate Incinerator. Processes, 2025, 13(12): 3928. https://doi.org/10.3390/pr13123928
9. Li Liu, Jiahe Sun, Ye Shui Zhang, Tanzila Anjum, **Dongkuan Zhang**, Junchao Yang, Jingliang Dong, Shaokoon Cheng, Guozhao Ji. Numerical Investigation of Coal and Rice Husk Co-Combustion in an Industrial-Scale Circulating Fluidized Bed: Hydrodynamics, Temperature, and Pollutant Emissions. Processes, 2026, 14(13): 2189. https://doi.org/10.3390/pr14132189
10. Tanzila Anjum, Tian Heng Qin, Gianni Olguin, Yinxiang Wang, **Dongkuan Zhang**, Xiaonan Kou, David K. Wang, Xiaozhen Zhang, Bradley Ladewig, Asim Laeeq Khan, Guozhao Ji. Impact of freeze drying on pore size distribution of amorphous silica membranes derived from gas permeation activation energies. Separation and Purification Technology, 2026, 395: 137736. https://doi.org/10.1016/j.seppur.2026.137736

Patents:

1. Guozhao Ji, **Dongkuan Zhang**, Yanjun Yang, Jiacheng Dai, Hong Tian, Tong Ren. A self-cleaning dual-spray nozzle system. China Patent No. CN121972345A, 2026.
2. Guozhao Ji, **Dongkuan Zhang**. An external-mixing dual-fluid atomizing spray nozzle. China Patent No. CN121847357A, 2026.
3. Guozhao Ji, **Dongkuan Zhang**. A device for self-settling fly ash in a waste incinerator. China Patent No. CN223137888U, 2025.
4. Guozhao Ji, **Dongkuan Zhang**. A fluidized bed reactor based on a Tesla valve. China Patent No. CN117504737B, 2024.
5. Guozhao Ji, **Dongkuan Zhang**. A device, method and application for self-settling fly ash in a waste incinerator. China Patent No. CN118912512A, 2024.
6. Guozhao Ji, **Dongkuan Zhang**. A device, method and application for self-settling fly ash in a waste incinerator. China Patent No. CN117308105A, 2023.
7. Jianxin Xia, **Dongkuan Zhang**, Fangfang Luan, Huade Cao, Dingbang Wei, Lei Guan. A flow field design method for polymetallic nodule collectors. China Patent No. CN113946927B, 2022.
8. Jianxin Xia, Lei Guan, Fangfang Luan, Huade Cao, Dingbang Wei, **Dongkuan Zhang**. A parameter matching method for polymetallic nodule collectors. China Patent No. CN114329824A, 2022.

Software Copyrights:

1. Nozzle (Pipeline) Clogging Risk Inspector Software (NRMS) V1.0. Registration No. 2025SR1164194, Copyright Dalian University of Technology, 2025.
2. Dual Spray Gun Self-Cleaning Control System (DSGSCS) V1.0. Registration No. 2025SR1799417, Copyright Dalian University of Technology, 2025.

Standards:

1. Technical Requirements for Hydrogen Production from Waste Based on High-Temperature Pyrolysis and Gasification (T/CITS 242-2025). Member of Drafting Group, China Inspection and Testing Society, 2025.

TECHNICAL SKILLS

- ✓ - CFD Simulation: FLUENT, Star-CCM+, Multiphase Flow Modeling.
- ✓ - Programming & Tools: Python, MATLAB, Data Analysis, Result Visualization.
- ✓ - Research Techniques: Flow Field Analysis, Heat and Mass Transfer, Experimental Testing.
- ✓ - Languages: English (Fluent), Mandarin (Native).